TEXORA · COURSE NOTES · LECTURE SHEET

Lettering, Numbering and Heading

The handwriting of engineering — single-stroke letters, standard heights and proportions, the spacing rules, and the title block that signs every sheet.

ME-1102 · ENGINEERING DRAWING-I · TOPIC 1 · LEVEL 1 TERM 1

1. Why a whole topic on handwriting?

A drawing speaks in two languages: its lines describe shape, and its **lettering** — titles, dimensions, notes — describes everything the lines cannot. A beautiful drawing with careless lettering is a ruined drawing: dimensions must be readable at a glance, at arm's length, by a stranger, possibly through a photocopy of a photocopy. Engineering therefore standardised the handwriting itself. The standard style is the **single-stroke Gothic** letter: every letter formed with strokes of uniform thickness (as if drawn with a single stroke of the pencil), without decoration, serifs or shading. Letters are either **vertical** (upright) or **inclined** (sloping at about 75° to the horizontal, i.e. 15° from vertical); a sheet uses one style throughout. Capital letters are the rule for titles, headings and most notes; lower-case appears in sentence-style notes by some standards but capitals dominate engineering sheets.

2. Heights, proportions and the two lettering types

Lettering is specified by the **height h of the capital letters**, chosen from the standard series 2.5, 3.5, 5, 7, 10, 14, 20 mm. Everything else — lower-case height, line thickness, spacing — is a fixed fraction of h, defined in two standard grades (ISO 3098):

FEATURE	TYPE A	TYPE B
Stroke thickness d	h/14	h/10 (the commoner, bolder grade)
Lower-case height	(10/14) h	(7/10) h
Minimum line spacing	≈ (22/14) h… in practice about 1.5–1.6 h	≈ 1.5–1.6 h
Typical use	Fine work, small text	General drawing work — the course default

Letter **widths** vary by letter (an I is narrow, a W is nearly as wide as it is tall) while the proportions of each letter — its skeleton — are standard, most capitals occupying roughly 5–7 units of width on a grid

where height is 7–8 units for Type B proportions as taught in drawing courses. Practical height assignments on a course sheet: the **main title** in the title block 6–8 mm, sub-headings 4–5 mm, dimension figures and notes 3–3.5 mm, and nothing anywhere below 2.5 mm.

3. Numerals and fractions

Numerals follow the same single-stroke construction and the same height as capitals in the same line (dimension figures are usually 3–3.5 mm). Decimal fractions are written with the decimal point at mid-height clearly placed (3.5, never a low dot that reads as dirt on the sheet); common fractions in dimensioning are written with a horizontal bar (½ style stacked form) sized about 1.5–2 times the figure height overall, or in the inclined-bar form in notes. Mixed practice to avoid: never mix inclined and vertical numerals on one sheet, and never let figures touch the dimension line they sit above — a small, uniform gap is standard.

4. The working technique

- **Guidelines** are drawn first, always: a pair of light lines fixing the cap height (and a third for lower-case where used). Skilled lettering is guideline lettering; freehand-height lettering drifts within a single word.
- **Spacing** is judged by *area*, not by ruler distance: letters are spaced so the white areas between neighbours look equal — straight-sided letters (H, I) stand closer, round letters (O, C) and pointed letters (A, V, W) need compensating adjustments. Word spacing ≈ the width of one capital O; line spacing ≈ 1.5 × letter height.
- **Consistency** is the gradeable virtue: same height to a fraction of a millimetre, same inclination, same stroke weight (a sharp, hard pencil — 2H/H for lettering — turned as you work to keep the point conical).
- Lettering is done **after** the drawing is laid out but the title block is lettered with the greatest care of all — it is the sheet's signature.

5. The heading: the title block

Every engineering sheet carries a **title block** — the standard heading — in the **bottom-right corner**, inside the border, a common college format being a rectangle about 185 mm × 65 mm divided into compartments. Its contents, lettered in the sizes above:

COMPARTMENT	CONTENT
Institution	Name of the institution (here, the institute/university) — often the largest lettering after the title

Title	The name of the drawing (e.g. “DEVELOPMENT OF A CONE”), 6–8 mm capitals
Drawn by	Student’s name, class/roll number
Scale	The scale of the drawing (e.g. 1:1, 1:2) or “NTS” (not to scale)
Date & sheet no.	Date of drawing; sheet number (“Sheet 3 of 8”)
Checked / remarks	Teacher’s signature column, and any revision or tolerance notes

Above the title block or across the sheet, a drawing may also carry a general **heading** for the exercise (e.g. “ORTHOGRAPHIC PROJECTION” as a sheet title at top centre in larger letters); the rule that unifies both is hierarchy — the eye should meet the most important words largest and first.

6. The textile thread

The drawing class is where an engineer’s sheets stop being homework and start being documents. Machine-layout drawings for a spinning or dyeing line, development patterns for the workshop, and the course’s own projection sheets are read by people who never meet their author — exactly the audience lettering standards were invented for. In industry the same discipline survives inside CAD title blocks and specification sheets, where the “lettering” is chosen from the same ISO series by whoever configures the template: the standard outlives the pencil.

7. Glossary — key terms of this topic

TERM	MEANING
Single-stroke Gothic	The standard letter style: uniform stroke thickness, no decoration.
Lettering height (h)	The height of capital letters, chosen from 2.5–20 mm standard series; all proportions derive from it.
Type A / Type B	The two standard proportion grades: stroke thickness h/14 (fine) and h/10 (general).
Guidelines	Light construction lines marking letter height before lettering begins.
Title block	The standard heading block at the bottom-right of a sheet: institution, title, drawer, scale, date, sheet number.
Inclined lettering	Letters sloping at about 75° to the horizontal.

8. Key points and summary

- Lettering is standardised communication: single-stroke Gothic, vertical or inclined (75°), one style per sheet, capitals dominant.
- Everything derives from capital height h (2.5–20 mm series): Type B stroke thickness h/10 is the general grade; guidelines and area-judged spacing produce uniformity.
- The title block at bottom-right — institution, title, drawn by, scale, date, sheet number — is the sheet's heading and signature.

9. Practice

Written and drawing exercises; brief model answers follow.

1. State the standard capital-height series, and assign heights for: a sheet's main title, a sub-heading, and dimension figures.
2. For Type B lettering of height 7 mm, compute the stroke thickness and the lower-case height.
3. List, in order, the compartments of a standard title block and the position of the block on the sheet.
4. Two students letter the word "DRAWING": one spaces letters by equal distances, the other by equal areas. Whose looks better and why?
5. Drawing exercise: rule guidelines for 7 mm Type B capitals and letter "ENGINEERING DRAWING" twice — once vertical, once inclined at 75° — keeping stroke weight uniform.
6. Drawing exercise: letter the numerals 0–9 and the dimension "Ø 25.50" at 3.5 mm height with correct decimal-point placement.
7. Find the faults: a sheet whose title is 3 mm, notes 7 mm, and title block at top-left. List every violation of this topic's standards.
8. Explain why lettering is done with a hard pencil and guidelines rather than freehand with a soft pencil.

Answer key

1. Series: 2.5, 3.5, 5, 7, 10, 14, 20 mm. Main title 6–8 mm; sub-heading 4–5 mm; dimension figures 3–3.5 mm.
2. Stroke thickness = h/10 = 0.7 mm; lower-case height = (7/10)h = 4.9 mm.
3. Institution; title of the drawing; drawn by (name/roll); scale; date; sheet number; checked/remarks. Position: bottom-right corner, inside the border.
4. The area-spaced one: letters have different shapes, so equal gaps make pointed/round letters look too far apart; equalising the white areas gives visually uniform spacing.

5. Self-check against §4: guidelines at 7 mm (and 4.9 mm if lower-case is included), uniform stroke, consistent 75° slope in the second version, word gap ≈ one O width.
6. Self-check: figures at uniform 3.5 mm; the decimal point placed clearly at mid-height, not on the baseline; the diameter symbol Ø drawn as a slashed circle of figure height.
7. Title smaller than notes reverses the required hierarchy (title should be largest, 6–8 mm; notes 3–3.5 mm); title block must be at bottom-right, not top-left.
8. A hard pencil (2H/H) holds a fine conical point, giving the uniform thin stroke the style requires and light guidelines that erase cleanly; a soft pencil's thick, fast-wearing point makes strokes vary in weight and smudges. Guidelines alone guarantee uniform height.
